# Anthony Hu

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## EDUCATION

**University of California, San Diego** — **La Jolla, California**
*B.S. in Mathematics and Computer Science* — *Expected June 2028*

- **GPA: 3.94 / 4.00** | **Provost Honors** (4 quarters) | **Coursework:** ML, Algorithms, Linear Algebra, Probability
- **Internship availability:** June 15 - September 25, 2027

## PUBLICATIONS & SUBMISSIONS

- Y. Zhang, Y. Tian, G. Chen, **A. L. Hu**, et al. "EgoPVA: When and Where Must We Look? Benchmarking Precise Visual Attention on Egocentric AI Glasses." Under review at CVPR 2027.
- Y. Zhao, **A. L. Hu**, Z. Huang, et al. "RoboClock: A Unified Benchmark for Time-Aware Robotic Intelligence." Under review at CVPR 2027.
- Y. Zhang, **A. L. Hu**, Y. Xu, et al. "GazeField: Fast In-Session Eye-Tracking Realignment for AI Glasses via Natural Head Motion and a Personalized Gaze Template." Under review at ACM CHI 2027.
- L. Wang, M. Sheng, L. Chen, ..., **A. L. Hu**, ..., X. Zheng. "Sub-Nanogram Resolution Measurement of Inertial Mass and Density Using Magnetic-Field-Guided Bubble Microthruster." Advanced Science, 2024, 11, 2403867.

## EXPERIENCE

**JD.com (NASDAQ: JD), JD Future Academy, Multimodal Laboratory** — **Beijing, China**
*Machine Learning Research Intern* — *Jun 2026 - Sep 2026*

- **Independently developed and deployed SVT** (Semantic Video Transition Detector) to clean the **~1.1B-video** pretraining corpus of **JoyAI-Video2**, a **200B** video generation model; cut false cuts **~82%** at ~97% recall.
- Architected a **dual-stream** detector: fine-tuned **TransNetV2** plus frozen **DINOv2 ViT-B/14** fused by a lightweight **1D-CNN verifier**; raised gradual recall **76.6% to 88.1%** and interval IoU **0.195 to 0.560**.
- Built a **2,288-video production benchmark**: SVT reached **96.3% F1** vs. TransNetV2 (94.3%), AutoShot (94.2%) and OmniShotCut (94.5%); **selective knowledge distillation** and synthetic jump cuts added **+1.4 F1**.
- **Increased inference throughput 3.2x** via FP16, GPU batching and shared FFmpeg decoding; shipped a **FastAPI** service.

**UCSD Integrated Systems Neuroengineering Lab (PI: Prof. Gert Cauwenberghs)** — **San Diego, California**
*Undergraduate Researcher | EgoPVA (CVPR 2027) / GazeField (CHI 2027)* — *Jan 2026 - Nov 2026*

- **Co-built EgoPVA (220 recordings, 23.9 h)** of when and where AI-glasses wearers must look closely, syncing **Meta Aria** video, IMU and gaze with **mobile EEG**; a sensor-based detector proposes events for a **VLM** to verify, cutting VLM cost **75x**.
- Led post-training: **LoRA SFT** of **Qwen VLMs (2B-8B)** predicting the PVA label, **8 criteria** and target box; raised Qwen3-VL-8B accuracy from **67.2% to 87.2%** and targets within 5% of image width from **26% to 71%**; benchmarked **10 VLMs**.
- **GazeField: cut gaze-calibration setup 65% (72 to 25 s)** at 2.83-degree error over 20 participants by aligning offline templates with ~5 s of natural head motion, with no calibration screen.
- **GazeSAM: cut pupil-segmentation CPU latency 50.7x** by fine-tuning **FastSAM-s** with **INT8 PTQ** + **ONNX Runtime**.

**STABLE Lab, UCSD (PI: Prof. Jishen Zhao)** — **San Diego, California**
*Undergraduate Researcher | RoboClock (CVPR 2027) / Self-Improving Agents* — *Oct 2026 - Present*

- **RoboClock:** designed the **10 sports and game tasks** of a 60-task, 5-domain time-critical benchmark in **Isaac Sim** (dual-arm ARX X5), scoring timing beyond physical success, e.g., catching a stalled bench press within **0.25-0.65 s**.
- Benchmarked **8 SOTA VLA/WAM policies** (e.g., pi0.5): **at most 0.6% temporal success**, and **0%** with inference latency.
- **Self-improving (RSI) agents:** found RRSI and PrimeAgent consume **2.3x more tokens** than the base agent by benchmarking with **Qwen3.8-27B**; now targeting efficiency and verification reliability.

**Shang Data Lab, UCSD (PI: Prof. Jingbo Shang)** — **San Diego, California**
*Undergraduate Researcher | AI Safety Benchmark (with Meta)* — *Oct 2026 - Present*

- **Led** the design and construction of a **200-task** benchmark for **explicit vs. implicit overscope** in LLM agents, grounded in real incident reports, GitHub issues and forums across **terminal, coding and computer-use** environments.
- Scored with **deterministic code probes** (LLM judge auxiliary); evaluated **12 models** (incl. GPT-6 Astra, Claude Fable, DeepSeek, Qwen, GLM, Nemotron) on overscope and task completion; **found overscope in frontier closed models**.

**London Stock Exchange Group (LSEG), Data & Analytics** — **Beijing, China**
*Software Engineer Intern* — *Jun 2025 - Sep 2025*

- **Cut Refinitiv pipeline runtime 40% (20 to 12 h)** for 38K+ securities via **async Python**; built a **Flask** SSH tool.

**Zhejiang University, Department of Engineering Mechanics** — **Hangzhou, China (Hybrid)**
*Research Assistant* — *Aug 2025 - Oct 2025*

- Improved **GCN/GIN** molecular-property models by **6-10% ROC-AUC**; cut training time **35%** with mixed precision.

## SKILLS

**Languages & Systems:** Python, C/C++, SQL, Bash, ONNX Runtime, FastAPI, FFmpeg, Docker, Isaac Sim, Git
**LLM & Multimodal:** PyTorch, Hugging Face (Transformers, PEFT), LLaMA-Factory, Qwen-VL, DINOv2, SAM
**ML Methods:** LoRA/SFT, multimodal data curation, benchmark design, distillation, INT8 quantization